Project Design Phase
Problem – Solution Fit Template

Date	07/03/2026
Team ID	SWTID-2026-6411
Project Name	product inventory management
Maximum Marks	2 Marks

Problem – Solution Fit Overview:
The Problem–Solution Fit ensures that the backend system is designed to solve real-world challenges related to product inventory management. It validates that the proposed API architecture, authentication system, and database design effectively address user needs for secure and efficient inventory handling.

Purpose:

- Address the lack of a secure and structured backend system for managing product inventories.
- Provide reliable REST APIs for product CRUD operations with private user-based access.
- Implement JWT-based authentication to ensure secure communication between clients and server.
- Enable efficient data storage and retrieval using MongoDB with proper authorization checks.
- Improve backend scalability and maintainability using modular Node.js and Express architecture.

Problem Statement:
Many small businesses and individual sellers rely on manual systems or unsecured applications to manage product inventories. Existing solutions often lack proper authentication, structured APIs, or user-specific data protection, which can lead to data inconsistency and unauthorized access. Without a well-designed backend architecture, managing inventory securely and efficiently becomes difficult.

Solution
A **Node.js and Express.js based Backend Application** designed to provide secure and scalable APIs for product inventory management.

- JWT authentication ensures that only authorized users can access protected routes.
- RESTful APIs allow users to create, read, update, and delete products efficiently.
- MongoDB database integration stores user and product data securely with userId-based authorization.
- Middleware-based architecture handles token verification, request validation, and error handling.
- Modular backend structure improves scalability, maintainability, and future feature expansion

Project Design Phase
Proposed Solution Template

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Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Many small businesses and individual users struggle to manage product inventory securely due to lack of structured backend systems. Existing solutions either lack proper authentication, have complex setups, or do not provide private user-based data management, leading to inefficient inventory handling.
2	**Idea / Solution Description**	Develop a Node.js and Express.js based backend system that provides secure REST APIs for product inventory management. The solution uses JWT authentication, MongoDB database integration, and middleware-based authorization to ensure users can securely perform CRUD operations on their own products.
3	**Novelty / Uniqueness**	The backend focuses on user-specific inventory management with JWT-protected APIs, modular MVC architecture, and scalable REST endpoints. It ensures data privacy by linking each product with a unique userId and implementing secure token verification for all protected routes.
4	**Social Impact / Customer Satisfaction**	The system helps small businesses and individual sellers manage inventory efficiently without expensive software. Secure backend APIs promote data privacy, reliability, and scalability, improving overall user trust and satisfaction.

Project Design Phase
Solution Architecture

Date	07/03/2026
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Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for the Product Inventory Management System focuses on building a secure, scalable, and maintainable backend using Node.js, Express.js, and MongoDB. The backend provides RESTful APIs for authentication and product management while ensuring data privacy through JWT-based authorization.

Goals of the Solution Architecture:

Identify the Best Tech Solution:

- Use Node.js, Express.js, MongoDB, and JWT authentication to build a secure and efficient backend API for product inventory management.

Define Structure & Characteristics:

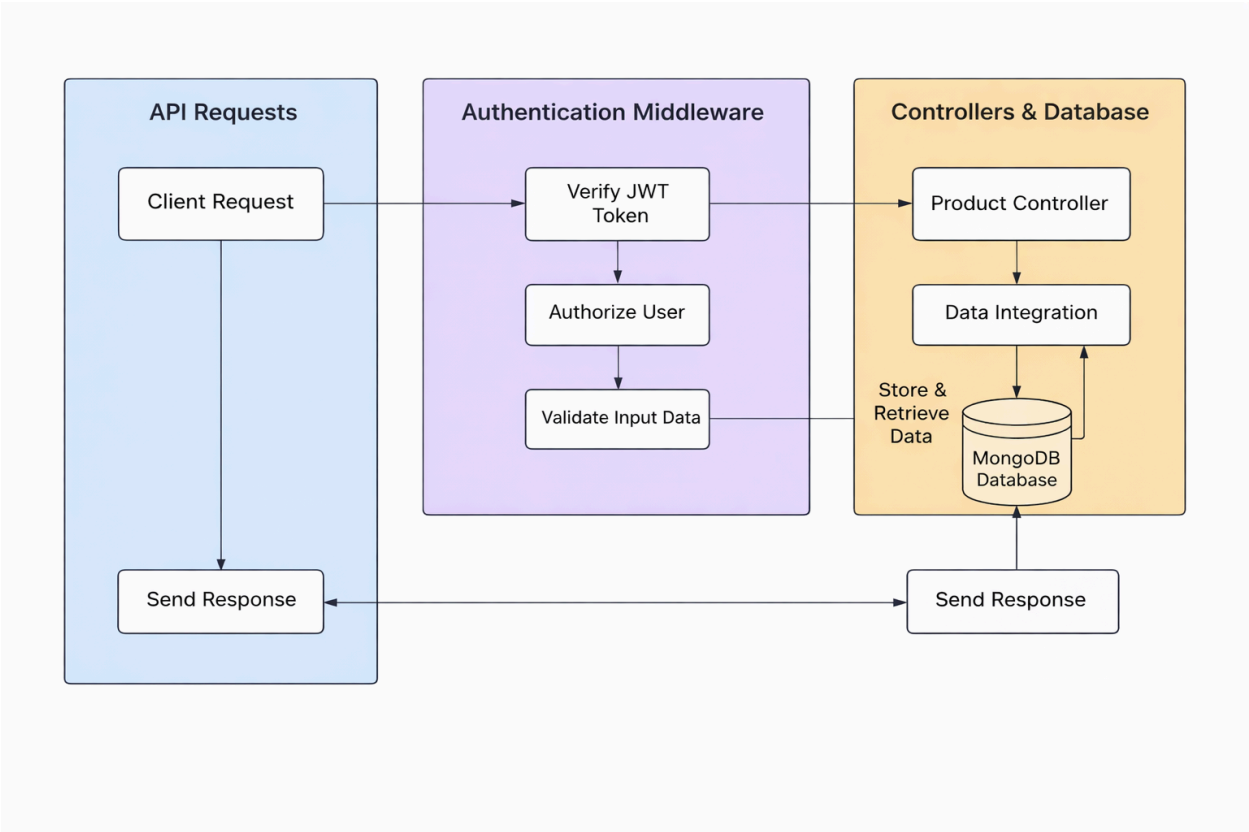
- Follow a modular MVC architecture to ensure scalability, maintainability, and clear separation of authentication and product management logic.

Outline Features & Development Phases:

- Plan development in phases including backend setup, authentication APIs, product CRUD operations, and testing for stable delivery.

Establish Specifications for Development & Delivery:

- Define REST API standards, token-based authorization, and structured middleware to maintain consistent and secure data flow.
- Optimize database queries and asynchronous API handling to provide fast response times and reliable backend performance.


API Requests
Client Request
Send Response
Authentication Middleware
Verify JWT Token
Authorize User
Validate Input Data
Controllers & Database
Product Controller
Data Integration
Store & Retrieve Data
MongoDB Database
Send Response